LEVEL 7 · PART III

Terraform / Infrastructure as Code

7

Goal: Define, version, review and safely change cloud infrastructure as code.

RECOMMENDED TIME	TOPICS
2-3 weeks	9

PRACTICE / CAPSTONE

Create a complete AWS environment with Terraform: VPC, subnets, security groups, EC2, ALB, RDS and Redis.

Topics

#	Topic	What you will learn
1	Providers and Resources **p.1405**	HCL, provider config, resource blocks, plan/apply/destroy workflow
2	Variables and Outputs **p.1419**	input types, validation, tfvars, locals, sensitive outputs
3	Modules **p.1431**	module structure, inputs/outputs, registry, versioning, composition
4	State and Remote Backends **p.1443**	what state is, S3 backend, security of state, import, moved blocks
5	Data Sources and Dependencies **p.1457**	data blocks, implicit/explicit depends_on, for_each/count, lifecycle
6	Reusable Environments **p.1471**	dev/stage/prod layout, workspaces vs directories, DRY patterns, Terragrunt mention
7	Terraform CI/CD **p.1485**	plan on PR, apply on merge, OIDC, policy checks, tflint/checkov
8	State Locking **p.1499**	DynamoDB/native S3 locking, stuck locks, force-unlock, concurrency
9	Drift Detection **p.1513**	manual changes, scheduled plan, detecting and reconciling drift
★	Level capstone project **p.1527**	Create a complete AWS environment with Terraform: VPC, subnets, security groups, EC2, ALB, RDS and Redis.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

